



**Verified Carbon
Standard**

SINOP LANDFILL GAS TO ELECTRICITY PROJECT

Document Prepared by Life İklim ve Enerji Ltd. Şti.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Sinop Landfill Gas to Electricity Project (hereafter referred to as the project) is located at Sinop Landfill Site in Hacıoglu Neighborhood, Central District, Sinop Province Turkey. The project is owned by Biokütle Enerji Elektrik Üretim A.Ş. (hereafter referred to as the project owner). The project is designed to utilize the gas produced in the existing landfill. The facility started operation on 29/06/2021¹ with half capacity of 1.453 MWm / 1.414 MWe. The total installed capacity will be 2.906 MWm / 2.828 MWe (2x1.453) MWm / (2x1.414) MWe, consisting of 2 sets of generators. The feed-in electricity is estimated to be 21,210 MWh / year² at full capacity during the 7-year crediting period.

The project began operation in 2021. It will handle an average of 174³ tons MSW per day.

Prior to the implementation of the project, the landfill site did not have gas extraction and utilization systems installed, and the LPG was directly emitted to the atmosphere. The project collects and utilizes LPG generated by Sinop landfill for power generation. Therefore, the project drastically reduces methane emissions, and at the same time, electricity is generated from the collected gas. Electricity generated is provided to the grid and the facility utilizes electricity generated by the project instead of drawing electricity from the grid.

The environmental, socio-economic, and technological benefits of the Project are described as follows:

- Reduction in fossil fuel use (imported or local) by using renewable energy resources,
- Reduction in greenhouse gas emissions from the landfill area by using biogas for electricity production,
- Reduction in energy production costs and imported energy amounts,
- Improvement of environmental conditions (GHG and odour) and safety in the landfill area.

The project is expected to reduce 30,789 tCO₂e in the first 7-year crediting period between commission date of 29/06/2022 and ending date of 28/06/2028.

¹ Sinop Landfill Provisional Acceptance

² EMRA Generation License

³ Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

Table 1 Milestones of the project activity

Date	Action
12.2020	Feasibility Report
11.12.2020	Operation Contract Between Project Owner and Sinop Coastal Municipalities Union
16.06.2021	EIA Exemption Letter
24.06.2021	EMRA Generation License
25.06.2021	Grid Connection Agreement
29.06.2021	Sinop Landfill Provisional Acceptance
18.10.2022	Local Stakeholder Consultation Meeting

1.2 Sectoral Scope and Project Type

Scope 1: energy industries (renewable-/non-renewable sources)

Scope 13: Waste handling and disposal.

Methodology:

AMS-III.G. Landfill methane recovery.

AMS-I.D Grid-connected renewable electricity generation

The project is not a grouped project.

1.3 Project Eligibility

The project is an LFG power generation project, utilizing landfill gas (LFG), which consists mainly of methane, for electricity generation, which is eligible under the scope of the VCS Program.

1.4 Project Design

☒ *The project includes a single location or installation only*

☐ *The project includes multiple locations or project activity instances, but is not being developed as a grouped project*

☐ *The project is a grouped project*

1.5 Project Proponent

Organization name	Biokütle Enerji Elektrik Üretim Anonim Şirketi
Contact person	Savaş Uzun
Title	CEO
Address	Soğuksu Mah. 308 Sok. Adabel Sit. No:42/5 Muratpaşa, Antalya
Telephone	+90 (242) 237 31 00
Email	savas.uzun@biosenerji.com

1.6 Other Entities Involved in the Project

Organization name	Life İklim ve Enerji Ltd. Şti.	
Role in the project	Carbon Certification Consultant	
Contact person	Utku Yazıcı	
Title	Expert	
Address	Oğuzlar Mah. 1377. Sk. No:19/9 06520, Balgat, Çankaya/Ankara	
Telephone	+90 312 481 2142	
Email	utku.yazici@lifeenerji.com	

1.7 Ownership

Biokütle Enerji Elektrik Üretim Anonim Şirketi holds all licenses, permits, and land use rights related to Sinop LFG and has obtained and owns all equipment used for its implementation.

1.8 Project Start Date

As it is stated in the Sinop Landfill Provisional Acceptance the start date of the project is 29/06/2021⁴.

⁴ Sinop Landfill Provisional Acceptance

1.9 Project Crediting Period

Crediting Period Type: Renewable (Renewable twice for a total of 21 years)

Crediting Period Start date: 29/06/2021

Crediting Period End date: 28/06/2028

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2021	11,789
2022	25,975
2023	28,169
2024	30,196
2025	32,039
2026	33,726
2027	35,276
2028	18,355
Total estimated ERs	215,525
Total number of crediting years	7
Average annual ERs	30,789

1.11 Description of the Project Activity

The Project, which is located in the Merkez District of Sinop, utilizes the landfill gas (LFG) generated by the existing landfill for energy generation. The Project has an operation period of 20 years under the current Project Owner. The Project Owner will then transfer the rights of operation of the Project to the Sinop Municipality at the end of the period.

Waste projections conducted under feasibility studies⁵ indicate that the landfill receives 173.6 t/day waste on average and the waste quantity is expected to rise during operation period from 155 t/day in 2021 to 236 t/day in 2040.

Waste is collected and deposited to the landfill by the Municipality. Then, landfill gas is extracted via landfill gas collection system. Collected gas is flared and utilized for electricity generation.

Steps in the landfill gas to electricity project:

- 1- Storage of waste in landfill
- 2- Generation of LFG at the landfill
- 3- Collection of LFG via pipelines
- 4- Booster system that draws collected LFG
- 5- Treatment of LFG via Chiller Unit
- 6- Electricity and thermal energy generation in gas engines
- 7- Electricity feeding to national power grid
- 8- Flaring process of extra gas that is not burnt in the engines.

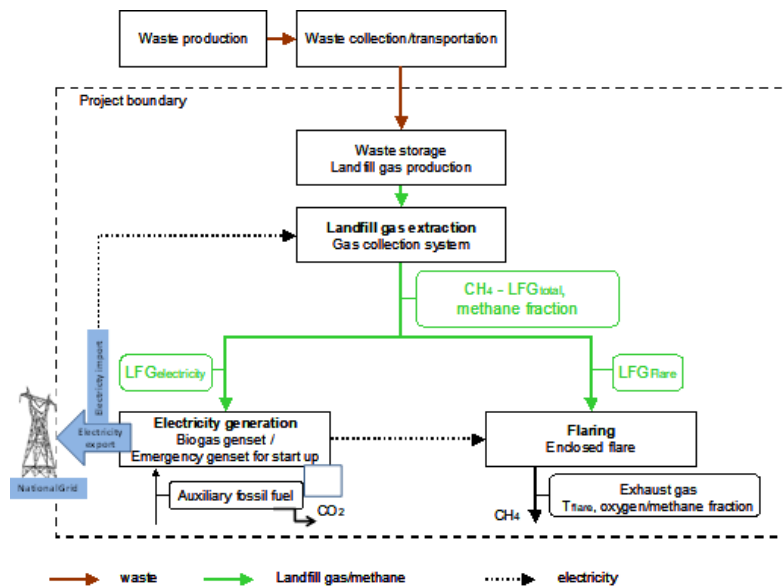


Figure 1 Boundary and the process flow of the Project

⁵ Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

The collection system which consists of pipelines and booster systems extracts gas from landfill and transmits LFG to chiller unit.

Chiller unit treats LFG and removes impurities and moisture from the gas. LFG is then transmitted to gensets.



Figure 2 Ürker Chiller UC ASH 15.1⁶

Table 2 Chiller technical specifications

Brand	Ürker Chillers
Model	UC ASH 15.1
Compressor Type	Semi-hermetic
Production year	2021
Chilling capacity	37.24 kW/h
Total power drawn	13.58 kW/h

Gensets generates electricity which is then fed to the grid.

The Project will utilize 2 units of Jenbacher brand gensets with a capacity of 1.414 MWe each. Currently only one unit is in operation and the second unit is planned to start operation in 2022 November⁷. The Project will have the capacity to generate 2.828 MWe when it reaches full capacity.

⁶ <https://www.urkerchillers.com/tr/img/pdf/Hava-Sogutmalı-Chiller-Katalogu.pdf>

⁷ Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

Technical Description

Genset

JGS 420 GS-L.L

static Grid Code

Sinop LF-2020-1



Electrical output 1414 kW el.

Emission values

NOx < 500 mg/Nm³ (5% O₂) | < 190 mg/Nm³ (15% O₂)

Figure 3 Jenbacher Generator Technical Specifications

0.01 Technical Data (at genset)

			100%	75%	50%
Power input	[2]	kW	3.403	2.625	1.848
Gas volume	*)	Nm ³ /h	851	656	462
Mechanical output	[1]	kW	1.453	1.089	727
Electrical output	[4]	kW el.	1.414	1.061	706
Heat to be dissipated (calculated with Glykol 37%)					
~ Intercooler 1st stage (Engine jacket water cooling circuit)	[9]	kW	291		
~ Intercooler 2nd stage (Low temperature circuit)		kW	75		
~ Lube oil (Engine jacket water cooling circuit)		kW	201		
~ Jacket water		kW	383		
~ Surface heat	ca. [7]	kW	122		
Spec. fuel consumption of engine electric	[2]	kWh/kWel.h	2,41	2,48	2,62
Spec. fuel consumption of engine	[2]	kWh/kWh	2,34	2,41	2,54
Lube oil consumption	ca. [3]	kg/h	0,29	~	~
Electrical efficiency	up to 35°C T1		41,5%	40,4%	38,2%

Figure 4 Jenbacher Generator Technical Specifications⁸

The total annual power generation is estimated to be 21,210 MWh/year. The expected average annual emission reductions are 30,789 tCO₂e in the first 7-year crediting period. The total estimated emission reductions over the crediting period of the project will be 215,525 tCO₂e. The milestone table of the project is presented below.

⁸ <https://www.innio.com/en/jenbacher/gas-engines/type-4/j420>

1.12 Project Location

The project is in Hacıoglu Neighborhood, which is located in Merkez District of Sinop Province in Turkey. The GPS coordinates of the Project area are provided in the following table.

Table 3 Project Area GPS coordinates

Corner No	E	N
1	668839.361	4646334.855
2	668859.503	4646335.610
3	668859.868	4646326.952
4	668865.364	4646327.145
5	668866.378	4646298.198
6	668840.680	4646297.217

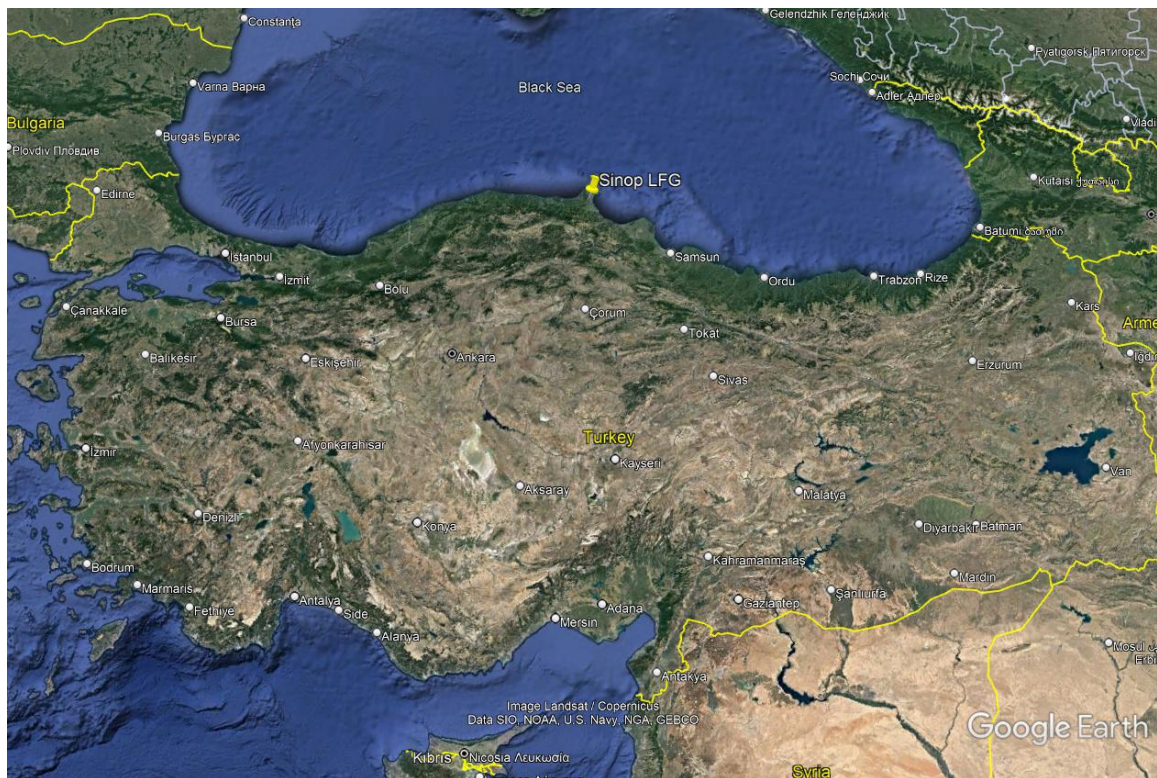


Figure 5 Location of the Project



Figure 6 Google Earth view of the Project and Landfill Site

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the implementation of the Project is the same as the baseline scenario described in Section 3.4.

- Prior to the implementation of the project, the landfill site did not have gas extraction and utilization systems installed.
- Gas generated due to decaying organic matter was released directly to the atmosphere.
- Equivalent electricity generated by the project was supplied by the grid, which is dominated by fossil fuel-based power plants.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project owner acquired all necessary legal permits prior to the commission of the facility. Most important legal documents are listed.

Project owner signed a contract⁹ with Union of Sinop Coastal Municipalities for operation of landfill and electricity generation from landfill gas on 11/11/2020.

⁹ Contract For the Operation of Landfill Belonging to The Union of Sinop Coastal Municipalities and the Generation of Electrical Energy from The Landfill Gas

The project has received an Environment Impact Assessment Exemption¹⁰ from Sinop Provincial Directorate of Environment and Urbanization on 16/06/2021.

Project has received the Electricity Generation License (EGL) on 24/06/2021¹¹.

Legal Framework

Laws and regulation concerning electricity generation and landfill operation are listed below. The regulations on waste management require precautions to prevent the explosion of landfill gas but do not require recovery or destruction.

The Project complies with all applicable laws and regulations.

Related Laws

- Electricity Market Law¹² (Law Number: 6446, ratified 14/03/2013, enacted 30/03/2013)

The Law aims to ensure the development of a financially sound and transparent electricity market operating in a competitive environment under civil law provisions and the delivery of sufficient, good quality, low cost and environment-friendly electricity to consumers and to ensure the autonomous regulation and supervision of this market.

No regulatory requirement for destruction of landfill gas.

- Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electricity Energy¹³ (Law Number: 5346, ratified 10/05/2005, enacted 11/08/1983)

The purpose of this Law is to expand the utilization of renewable energy resources for generating electrical energy, to benefit from these resources in a secure, economical and qualified manner, to increase the diversification of energy resources, to reduce greenhouse gas emissions, to assess waste products, to protect the environment and to develop the related manufacturing sector for realizing these objectives.

The Law brings an incentive of 13.3 \$ cent/kWh for the electricity production from biomass. It also brings incentives for local equipment purchases such as turbines, engines, cogeneration

¹⁰ Environment Impact Assessment Exemption Letter

¹¹ Electricity Generation License

¹² <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4628-20080709.pdf>

¹³ <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.5346.pdf>

- Environment Law¹⁴ (Law Number: 2872, ratified 09/08/1983, enacted 11/08/1983)

This Law addresses the ecological security of the population, the rational use of natural resources, nature conservation and environmental protection. Additional Article 6 says that clean air policies should be applied in provinces and districts, and air quality should be monitored. Methodologies for determination, monitoring and measurement of air quality, air quality limit values, precautions to prevent air pollution and public awareness are responsibilities of the Ministry of Environment and Forestry.

No regulatory requirement for destruction of landfill gas.

Related Regulations

- Regulation on Solid Waste Management¹⁵ dated 02/04/2015 and numbered 29314

The regulation aims to determine general waste management principles to protect the environment and human health from generation to disposal of waste.

No regulatory requirement for destruction of landfill gas.

- “Regulation on landfilling¹⁶” dated 26/03/2010 and numbered 27533

The regulation aims to protect the environment by minimizing the negative impacts of leachate and landfill gas on soil, air, underground and surface water

No regulatory requirement destruction of landfill gas.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The Project has not been registered or is seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

Since the Project did not apply for any other programs, it did not get rejection from any either.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

¹⁴ <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=2872&MevzuatTur=1&MevzuatTertip=5>

¹⁵ https://webdosya.csb.gov.tr/db/cygm/editordosya/Atik_Yonetimi_Yonetmeligi-2017.docx

¹⁶ <http://www.mevzuat.gov.tr/Metin.Aspx?MevzuatKod=7.5.13887&MevzuatIliski=0&sourceXmlSearch=>

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

Not applicable. The project activity is neither included in an emissions trading program nor does it take place in a jurisdiction or sector in which has binding limits on GHG emissions.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

The project has not sought or received another form of GHG-related environmental credit or renewable energy certificates.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project not only aims to generate energy from a sustainable and renewable energy source but also to find a sustainable solution to an unmanaged landfill which released landfill gas to atmosphere prior to the installation of facility. The Project will contribute to long term sustainable development of the region.

Main contributions to the Sustainable Development Goals¹⁷ adopted by the United Nations are summarized.

SDG 7: Affordable and Clean Energy

- The project converted previously unmanaged landfill into a biogas-to-energy facility which is also a clean and efficient facility for waste disposal and handling.
- Generates energy from a renewable source.
- Provides energy to the grid from renewable a source and replaces energy generated by fossil fuel utilization.

SDG 6: Water Quality

- Replaces electricity supplied to grid by fossil fuel consuming power plants that use water for cooling purposes, and thus, reduces water consumption in the process.

¹⁷ <https://www.undp.org/sustainable-development-goals>

SDG 8: Decent Work and Economic Growth

- Creates employment opportunities related to the management, operation and maintenance of landfill and the methane recovery and electricity generation systems
- Turkey is an energy and fuel dependent country, and the project contributes to reduction of energy dependency for the country and thus contributes to economic growth.

SDG 13: Climate Action

- Management of unmanaged landfill site by establishing a new, clean and efficient facility for waste disposal and handling.
- Collection of landfill gas, reduction and prevention of direct emission of GHG to the atmosphere. Estimations indicate that The Project can reach an average emission reduction of 30,789 tCO₂eq annually.

1.17.2 Sustainable Development Contributions Activity Monitoring

The project, located in Sinop Turkey, reduces GHG emissions by capturing and utilizing methane generated in the landfill. Project prevents direct methane emissions from the landfill to the atmosphere via collection system and generates renewable energy via flaring methane and by doing so increases renewable energy share in the Turkish electricity grid. Reduces amount of wastewater generated by displacing energy generated by fossil fuel plants that use water for cooling. The project also creates equal and safe employment opportunities in the region.

Table 4: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value."	Implemented activities to increase	The Project Activity will provide employment to at least 5 people.	Project Proponent provides employment to at least 5 people during the project lifetime complying with Labor Law in Turkey.
2)	8.8	Protect labour rights and promote safe working environments	Implemented activities to increase	Occupational Health and Safety training is provided to employees at least once a year	Occupational Health and Safety training is provided to employees at least once a year

3)	7.2	By 2030, increase substantially the share of renewable energy in the global energy mix	Implemented activities to increase	The annual electricity generation estimation is 21,210 MW. The generated electrical energy is directly fed to the Turkish national grid. Generated electricity is equal to the consumption of around 8,000 households.	The annual electricity generation estimation is 21,210 MW. The generated electrical energy is directly fed to the Turkish national grid. Generated electricity is equal to the consumption of around 8,000 households.
4)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	Since the project collects and utilizes LFG, it will eliminate around 30,789 tCO ₂ annually.	Project aims reducing carbon emission by 215,525 tons for the first crediting period. (7 years)
5)	6.3	By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally	Implemented activities to increase	Since it is renewable energy project, project activity does not consume any water for cooling purposes like fossil fuel energy plants. Therefore, this project provides significant water use avoidance and thus protects the environment. The project is estimated to avoid around 517,845 m ³ wastewater yearly at full capacity.	The project is estimated to avoid around 517,845 m ³ wastewater yearly at full capacity.

1.18 Additional Information Relevant to the Project

Leakage Management

In accordance with the methodologies applied to the project, no leakage effects are accounted for.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

N/A

2 SAFEGUARDS

2.1 No Net Harm

Project owner applied for an evaluation regarding Environmental Impact Assessment (EIA) Regulation to Sinop Provincial Directorate of Environment and Urbanization on 23/12/2020 and the Project received EIA Exemption after evaluation of the project from the Directorate on 16/06/2021.

The current practice on the site is a managed landfill with no methane recovery. The Project has several environmental benefits:

- It will cover the landfill area with soil, reducing the adverse visual impacts.
- The cover and an effective landfill gas collection system will reduce odour from the landfill
- The LFG recovery will reduce GHG emissions and help in tackling global warming
- It will set a good example for proper waste management with improved health and safety impacts
- It will minimize explosion and fire risks by decreasing gas accumulation
- It will feed clean energy to the grid, thereby reducing emissions from power generation

As one of the sustainable renewable energy generation technologies, the Project reduces the environmental impacts of the current practice on the landfill site considerably.

During the meeting a presentation was provided detailing the project, its grievance mechanisms, environmental and social impacts, contributions to sustainable development goals.

In addition, during the local stakeholder consultation meeting, participants filled in an evaluation form regarding the project to document their feedback on the impacts of the project. Attendance is recorded.

During the meeting, the stakeholders didn't raise any concerns about the project, and they indicated since the implementation of the project waste and odor issue caused by the wild dumpsites disappeared.

Grievance mechanism was established with the support of the Mukhtar for stakeholders to forward their requests or concerns regarding the project activities. In Turkey, mukhtars are the most relevant persons in the villages for establishing a grievance mechanism. The Mukhtar was provided all the contact information of the responsible persons from the project activity. Whenever a complaint arises from the local people or Mukhtar himself arises a concern regarding the project activity, he can directly contact the relevant responsible persons.

2.3 Environmental Impact

The environmental matters in Turkey are jurisdiction of the Ministry of Environment and Urbanization. The Law defines among others the general requirements of the Environmental Impact Assessment (EIA) states specific requirements according to the type of project, location, size, and other characteristics.

Project owner applied for an evaluation regarding Environmental Impact Assessment (EIA) Regulation to Sinop Provincial Directorate of Environment and Urbanization on 23/12/2020 and the Project received EIA Exemption after evaluation of the project from the Directorate on 16/06/2021.

2.4 Public Comments

Following initial project listing, the VCS/Verra project Sinop Landfill Gas to Electricity Project will be open for 30 days of public consultation.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Approved methodologies applied to the project are;

1. AMS-III.G. Landfill methane recovery Version 10.0

This methodology is described as *“Capture and combustion of methane from landfills used for disposal of residues from human activities including municipal, industrial and other solid wastes containing biodegradable organic matter”*¹⁸ in the CDM Methodology Booklet published by United Nations.

AMS-III.G Version 10 refers to following tools where applicable:

- "Emissions from solid waste disposal sites" version 08.0
- "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation " version 03.0
- “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion” Version 03.0
- “Project Emissions from Flaring” Version 03.0
- “Determining the baseline efficiency of thermal or electric energy generation systems” Version 03.0
- “Positive lists of technologies” Version 03.0

2. AMS-I.D Grid-connected renewable electricity generation Version 18.0

AMS-I.D Version 18 refers to following tools where applicable:

- "Tool to calculate the emission factor for an electricity system" version 07.0
- “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion” Version 03.0
- “Tool to determine the remaining lifetime of equipment” Version 01
- “Project and leakage emissions from biomass” Version 04.0

¹⁸ https://cdm.unfccc.int/methodologies/documentation/meth_booklet.pdf

- “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” Version 03.0.1

3.2 Applicability of Methodology

The baseline calculation for the Project follows the procedures for categories:

- Small-scale Methodology: Landfill methane recovery AMS-III.G / Version 10;

This methodology applies to Sinop landfill gas project activity, where the baseline scenario is: in the absence of the project activity, MSW is left to decay within the project boundary, and methane is emitted into the atmosphere. There was no legal requirement for capturing the methane emissions in the baseline scenario. Furthermore, for this project activity, it is estimated that annual emission reductions will be less than 60 kt CO₂e which is lower than the threshold value specified in the applied methodology. Also, in the absence of the project activity, organic waste would not be recycled and just left to decay. Moreover, in this project activity, the management of the solid waste disposal site wasn't deliberately changed to increase methane generation compared to the situation before the implementation of the project activity.

Applicability of Methodology	Justification
AMS-III.G., “Landfill methane recovery” version 10.0	
Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity.	The Project Activity is estimated to result in less than 60 kt CO ₂ equivalent annually, and 30 kt CO ₂ average at max capacity. So, this requirement applies to the Sinop LFG Project.
The proposed project activity does not reduce the quantity of organic waste that would have been recycled in the absence of the project activity.	Sinop LFG Project activity does not reduce the quantity of organic waste that would have been recycled in the absence of the project activity.
This methodology is not applicable if the management of the solid waste disposal site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement).	The management of the landfill site belongs to the Municipalities 'Union so that Project Proponent does not have any control over the SWDS management. Thus, the SWDS management is not deliberately worsened to increase methane generation prior to the project implementation.

The applicability conditions of the tools referred by the applied methodology AMS-III.G. are explained in the following table.

Applicability of Methodology	Justification
AMS-III.G., "Landfill methane recovery" version 10.0	
"Emissions from solid waste disposal sites" version 08.0	The Project Activity falls under Application A. project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring the methane. The tool is only applied for an ex-ante estimation of emissions in this report. The emissions will then be monitored during the crediting period.
"Tool to calculate project or leakage CO2 emissions from fossil fuel combustion" Version 03.0	Not applicable. Emissions referred to are negligible compared to the project emissions from electricity consumption or flaring.
"Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" Version 03.0"	This tool is applied to calculate the project emissions from electricity consumption by the project activity. Scenario, A of Paragraph 5 (a) of the tool, apply for this project activity. Electricity consumption is from the grid, and there are no captive power plants on site.
"Project Emissions from Flaring" Version 03.0	The Project Activity employs an enclosed flare unit, and Methane is the component with the highest concentration in the flammable residual gas, and the source of the residual gas is from a biogenic source (e.g. Landfill gas). Additionally, the residual gas has sufficient combustible gas present to sustain combustion. Moreover, there are operating specifications provided by the manufacturer of the flare. For this reason, the tool is applicable for this project activity.
"Determining the baseline efficiency of thermal or electric energy generation systems" Version 03.0	Not applicable. Turkish Ministry of Energy and Natural Resources publishes Emission factor applicable for the national grid for wind, solar and other renewable projects.
"Positive lists of technologies" Version 03.0	This tool is applied for the demonstration of the additionality of the Project Activity. According to Paragraph 5 of the tool, the project activity falls under the category of 5.1.1 Landfill gas recovery and its gainful use. The LFG is used to generate electricity in a power plant with a total capacity below 10MW.

Applicability of Methodology	Justification
AMS-I.D., "Grid-connected renewable electricity generation" version 18.0	
Install a Greenfield plant	The Project is a Greenfield project involving the setting up of a new power plant at a site where there were no power plants in the pre-project scenario. The Project is only 2.828 MWe.
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:	The Project involves the installation of a Greenfield plant at a site where there was no renewable power plant operating before the implementation of the Project.
If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The Project Activity doesn't have any non-renewable component. The Project involves electricity generation by utilizing LFG produced from a landfill with a total capacity of 2.828 MWe.
Combined heat and power (cogeneration) systems are not eligible under this category.	The Project Activity is not a Combined heat and power (cogeneration) system. The Project's recovered methane is used only for electricity generation and supplied to the grid. It is not utilized for heat generation or cogeneration.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the Project should be lower than 15 MW and should be physically distinct from the existing units.	The Project Activity does not involve the capacity addition of renewable energy generation units at an existing renewable power generation facility since there was no renewable energy generation in the project site prior to Sinop LFG.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated, or replacement power plant/unit shall not exceed the limit of 15 MW.	The Project Activity is not a retrofit, rehabilitation or replacement project.
In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane	The recovered methane is used for electricity generation and supplied to the grid. The baseline for the electricity component is in accordance with

emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with the procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration, other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	the procedure prescribed under this methodology. The recovered methane is not used for heat generation or cogeneration.
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	The methane is not sourced from dedicated plantations.

The applicability conditions of the tools referred by the applied methodology **AMS-I.D.** are explained in the following table.

Applicability of Methodology	Justification
AMS-I.D., "Grid-connected renewable electricity generation" version 18.0	
"Tool to calculate the emission factor for an electricity system" version 07.0	Not applicable. Turkish Ministry of Energy and Natural Resources publishes Emission factor applicable for the national grid for wind, solar and other renewable projects.
"Tool to calculate project or leakage CO2 emissions from fossil fuel combustion" Version 03.0	Not applicable. Emissions referred to are negligible compared to the project emissions from electricity consumption or flaring.
"Tool to determine the remaining lifetime of equipment" Version 01	This tool is not used for the determination of equipment lifetime. The Project is designed for an operational lifetime of 20 years, according to the contract with the Solid Waste Management Union of Sinop Coastal Municipalities.
"Project and leakage emissions from biomass" Version 04.0	Not applicable. Project Activity doesn't include biomass.
"Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period" Version 03.0.1	Not applicable. Project Activity is not at the renewal of the crediting period.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted for since the CO ₂ is also released under the project activity
		CH ₄	Yes	Major source of emissions in the baseline.
		N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from SWDS. This is conservative
		Other	No	Excluded for simplification. This is conservative
	Emissions from electricity generation	CO ₂	Yes	Major emission source if power generation is included in the project activity
		CH ₄	No	Excluded for simplification. This is conservative
		N ₂ O	No	Excluded for simplification. This is conservative
		Other		
Project	Emissions from electricity consumption due to the Project activity	CO ₂	Yes	May be an important emission source.
		CH ₄	No	excluded for simplification. This emission source is assumed to be very small
		N ₂ O	No	excluded for simplification. This emission source is assumed to be very small
		Other	No	Excluded for simplification. This is conservative
	LFG capture and flaring	CO ₂	No	It is part of the natural carbon cycle, where the carbon was previously sequestered from the atmosphere.
		CH ₄	Yes	When flare does not operate, methane emissions are subtracted.

Source	Gas	Included?	Justification/Explanation
	N ₂ O	No	Excluded for simplification. This is conservative.
	Other	No	Excluded for simplification. This is conservative.

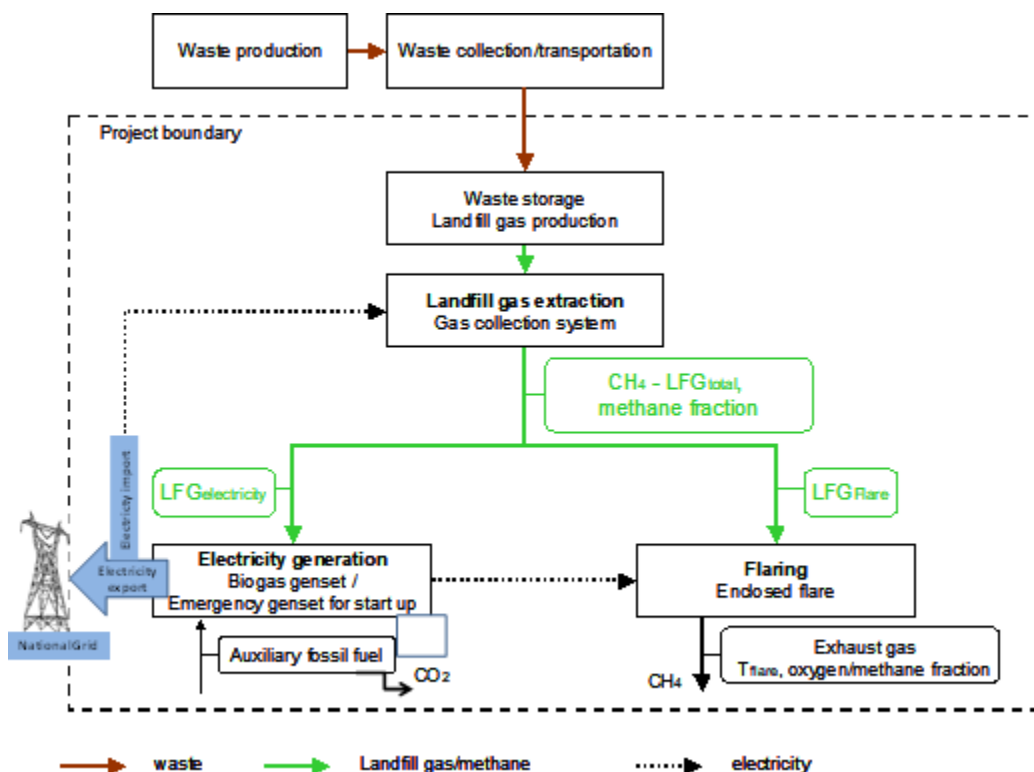


Figure 9 Boundary of the Project

3.4 Baseline Scenario

In the baseline scenario the biomass and organic matter decays in the landfill and methane is produced during the process. The produced methane is then released directly to the atmosphere without the proposed project activity. This scenario is identical to the baseline scenario described in **AMS-III.G**. There is no incentive or a national regulation/standard for prevention of methane emissions originating from a landfill. Baseline emission calculations covers the effects of methane oxidation difference of with and without project scenario.

In the absence of the project, scenario also complies with the baseline scenario described in **AMS-I.D**. In this scenario the electricity is provided to the Turkish national grid from power plants that mainly utilizes fossil fuels.

In Turkey there is no legal framework for abatement of emissions of greenhouse gases from landfill. In the absence of specific laws and regulations regarding GHG emissions there is no incentive for capturing and utilization of the LFG.

Therefore, as explained above, baseline emissions for the Project are:

- A. Landfill Gas: In the absence of the project activity, biomass and other organic matter are left to decay within the project boundary and methane is emitted into the atmosphere
- B. Electricity: Product of electricity energy baseline expressed in kWh of electricity produced by the renewable generating unit multiplied by an emission factor. Combined margin (CM) is adopted for emission factor.

The Project owner does not have fossil fuel-fired power plants or control of emissions resulting from the baseline scenario. The Project partially reduces LFG emissions from landfill and replace the electricity generation from grid-connected fossil fuel-fired thermal power plants, thereby reducing greenhouse gas emissions.

3.5 Additionality

The proposed Project is deemed additional, as defined by the applied approved methodologies.

AMS-III.G. Small Scale Methodology Landfill Methane Recovery (Version 10.0) states that project activities and PoAs at new and existing landfills are deemed additional automatically as long as in the baseline scenario LFG was only vented. Other criteria stated in the methodology also applies to the project; LFG is flared, and total electricity generation capacity has to be under 10MW, and the project capacity is 2.282 MW.

Therefore, the project is deemed automatically additional.

3.6 Methodology Deviations

Not applicable.

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions calculations are determined according to AMS-III.G and AMS-I.D methodologies. These emissions can be divided into two parts. First part is from the methane emissions originating from the SWDS and the second part originates from the electricity generation using fossil fuels or supplied by the grid (electricity displacement) in the absence of the project activity.

$$BE_y = BE_{CH_4,y} + BE_{electricity,y} \quad \text{(Equation-1)}$$

Where;

BE_y = Baseline emissions in year y (tCO_{2e}/y)

$BE_{CH_4,y}$ = Baseline emissions of methane from the SWDS in year y (tCO_{2e}/y)

$BE_{electricity,y}$ = Baseline emissions associated with electricity displacement y (tCO_{2e}/y)

Baseline Methane Emissions

The ex-ante emission reduction by the project activity will be calculated according to AMS-III.G with the following equation:

$$BE_y = \eta_{PJ} \times BE_{CH_4,SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} + GWP_{CH_4} \quad \text{(Equation-2)}$$

Where;

$BE_{CH_4,SWDS,y}$ = Methane emission potential of a solid waste disposal site (tCO_{2e})

OX = Oxidation factor (A default value of 0.1 is used)

$F_{CH_4,BL,y}$ = Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (tCH₄)

η_{PJ} = Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50% is used

GWP_{CH_4} = Global Warming Potential for methane

$F_{CH_4,BL,y}$ is taken as zero since there is no regulation in Turkey that advises the capture of methane emissions. ($F_{CH_4,BL,y} = 0$)

$BE_{CH_4,SWDS,y}$ is calculated according to Tool 04 Emissions from solid waste disposal sites Version 08.0.

$$\left. \begin{array}{l} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{array} \right\} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \\ \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j})) \quad \text{(Equation-3)}$$

Where;

- BE_{CH₄,SWDS,y}** = Methane emission potential of a solid waste disposal site (tCO_{2e})
- x** = Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)
- y** = Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)
- DOC_{f,y}** = Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
- W_{j,x}** = Quantity of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
- φ_y** = Model correction factor to account for model uncertainties for year y
- f_y** = Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
- GWP_{CH₄}** = Global Warming Potential for methane
- OX** = Oxidation factor (A default value of 0.1 is used)
- F** = Fraction of methane in the SWDS gas (volume fraction)
- DOC_j** = Fraction of degradable organic carbon in the waste type j (weight fraction)
- k** = Decay rate for the waste type j (1 / yr)
- j** = Type of residual waste or types of waste in the MSW

The project gas collection efficiency is 85% according to the feasibility report.

Baseline Emissions From Electricity Displacement

According to AMS-I.D, baseline emissions include only CO₂ emissions from electricity generation in power plants displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_{y,electricity} = EG_{PJ,y} + EF_{grid,y} \quad \text{(Equation-4)}$$

Where;

- BE_{electricity,y}** = Baseline emissions associated with electricity displacement y (tCO_{2e}/y)
- EG_{PJ,y}** = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO2 emission factor of grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO2/MWh)

$$EG_{PJ} = EG_{PJ, facility, y} \quad \text{(Equation-5)}$$

Where;

$EG_{PJ, facility, y}$ = Quantity of net electricity generation that is produced by the project plant to the grid in year y (MWh)

Determination of the emission factor for electricity generation ($EF_{grid,y}$)

To find $EF_{grid,y}$, “Tool to calculate the emission factor for an electricity system” (V 07.0) is applied to calculate combined margin. The following six steps below, provided in the Methodological Tool, are used to calculate the combined margin (CM) emission factor:

For the emission factors for electricity generation, the publication of the Turkish Ministry of Energy and Natural Resources, which indicates Turkey’s National Electric Grid Emission Factor for 2022¹⁹, was used. The publication includes calculated Emission Factor values that are Operating Margin (OM), Growth Based Margin (Build Margin-BM) and Combined Margin (CM) Emission Factors, for the relevant year with the usage of the IPCC’s Clean Development Methodology Tool 07-V06.0. For this calculation, information regarding the data set employed is given below in detail;

- TEİAŞ Turkey's electricity generation-consumption and loss statistics,
- Commonly prepared report under Turkey's National Greenhouse Gas Inventory Reporting Format. - Common Reporting Format (CRF) tables for electricity generation (1.A.1.a.i) emission values
- Chronological order of power generation plants from TEİAŞ Load Dispatch Department with commissioning dates, plant names, fuel types, installed power values, electricity generation for the calculated year
- Checking off Volunteers from the websites of Gold Standard (GS) and Verified Carbon Standard (VCS) for the ownership status of the carbon reduction certificate and,
- From Clean Development Mechanism (CDM) Tool 009- V2.0, Power plant efficiency figures are used

According to this publication;

- Operating Margin-OM; 0.7424 tCO2/MWh
- Build Margin-BM; 0.3680 tCO2/MWh

¹⁹

<https://enerji.gov.tr/Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

Combined Margin-CM3 (for other renewable than solar and wind); 0.5552²⁰ tCO₂/MWh

4.2 Project Emissions

Project emissions are calculated as follows:

$$PE_y = PE_{EC,y} + PE_{FC,y} + PE_{DT,y} + PE_{SP,y} \quad \text{(Equation-6)}$$

Where:

PE_y = Project emissions in year y (t CO₂/yr)

PEEC,_y = Emissions from consumption of electricity due to the project activity in year y (t CO₂/yr)

PEFC,_y = Emissions from consumption of fossil fuels due to the project activity, for purpose other than electricity generation, in year y (t CO₂/yr)

PEDT,_y = Emissions from the distribution of compressed/liquefied LFG using trucks, in year y (t CO₂/yr)

PESP,_y = Emissions from the supply of LFG to consumers through a dedicated pipeline, in year y (t CO₂/yr)

The project is only included the emissions from consumption of electricity, not involved in the emissions from consumption of fossil fuels, the distribution of compressed/liquefied LFG using trucks and the supply of LFG to consumers through a dedicated pipeline, thus,

$$PE_y = PE_{EC,y} \quad \text{(Equation-7)}$$

The project emissions from consumption of electricity by the project activity (PEEC,_y) shall be calculated using the methodological tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”. When applying the tool:

- (a) ECPJ,_{j,y} in the tool is equivalent to the quantity of electricity consumed by the project activity in year y (EGEC,_y); and
- (b) If in the baseline a proportion of LFG is destroyed (FCH₄,BL,_y>0), then the electricity consumption in the tool (ECPJ,_{j,y}) should refer to the net quantity of electricity consumption (i.e. the increase due to the project activity). The determination of the quantity of electricity consumed in the baseline shall be transparently documented in the PD.

The calculation equation is as follows:

²⁰ <https://enerji.gov.tr//Media/Dizin/EVCED/tr/ÇevreVeİklim/İklimDeğişikliği/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EL,j,y} \times (1 + TDL_{j,y}) \quad \text{(Equation-8)}$$

Where:

PEEC, y = Project emissions from electricity consumption in year y (t CO₂ / yr)

ECPJ, j, y = Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)

EFEL, j, y = Emission factor for electricity generation for source j in year y (t CO₂/MWh)

TDLj, y = Average technical transmission and distribution losses for providing electricity to source j in year y

j = Sources of electricity consumption in the project

As stated above, according to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”(version 03.0), for the project is in the case of Scenario A: Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the “Tool to calculate the emission factor for an electricity system” (version 07.0) (EFEL,j,y = EFgrid,CM,y).

According to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0), the project belongs to the case of Scenario A, use as default values of 20% for project emission, i.e ,TDLj, y =20%.

ECPJ, j, y is ex ante determined as 0 in the PD and will be monitored ex post in the verification period.

4.3 Leakage

According to paragraph 20 of AMS-III.G leakage must be considered if the methane recovery technology is transferred from another activity. As this project doesn't involve transfer of any equipment from another activity therefore the leakage has been considered zero.

Following paragraph 42 of AMS-I.D general guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues. Biomass residues are defined as per CDM glossary of Terms version 09.1 as “non-fossilized and biodegradable organic material originating from plants, animals and micro-organisms which is a by-product, residue or waste stream from agriculture, forestry and related industries”. This project doesn't pertain to the use of biomass residues. Therefore, no leakage effects need to be accounted for under the AMS-I.D.

4.4 Estimated Net GHG Emission Reductions and Removals

The emission reduction achieved by the project activity is constituted by both the methane destruction and grid displacement components.

$$ER_y = BE_y - PE_y - LE_y \quad \text{(Equation-9)}$$

Where:

ER_y = Emission reductions in project year y (t CO₂)

BE_y = Baseline emissions in project year y (t CO₂)

PE_y = Project emissions in project year y (t CO₂)

LE_y = Leakage emissions in project year y (t CO₂)

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2021	12,770	981	0	11,789
2022	28,201	2,226	0	25,975
2023	30,639	2,470	0	28,169
2024	32,891	2,695	0	30,196
2025	34,939	2,900	0	32,039
2026	36,813	3,087	0	33,726
2027	38,536	3,260	0	35,276
2028	20,065	1,710	0	18,355
Total	234,854	19,329	0	215,525
Average	33,551	2,761	0	30,789

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	OX
Data unit	Dimensionless
Description	Fraction of methane that would be oxidized in the top layer of the SWDS in the baseline
Source of data	Consistent with how oxidation is accounted for in the methodological tool “Emissions from solid waste disposal sites”
Value applied:	0.1
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Baseline emission calculation
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	t CO2e/t CH4
Description	Global warming potential of CH4
Source of data	IPCC AR5 ²¹
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value of 28 from IPCC fifth Assessment Report (AR5). Shall be updated according to any future COP/MOP decisions.
Purpose of Data	Ex ante calculation of baseline emissions
Comments	-

²¹ <https://www.ipcc.ch/report/ar5/syr/>

Data / Parameter	η_{PJ}
Data unit	Dimensionless
Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	Feasibility study ²²
Value applied	85
Justification of choice of data or description of measurement methods and procedures applied	The collection system which will be installed on the Project site will be able to collect a certain fraction of the total LFG generation. This collection ratio is affected principally by the Project's technical design, therefore, the value suggested in the feasibility study is used.
Purpose of Data	Ex Ante Baseline emission calculation
Comments	-

Data / Parameter	ϕ_y
Data unit	-
Description	Default value for the model correction factor to account for model uncertainties
Source of data	"Methodological Tool: Emissions from solid waste disposal sites" (Version 08.0)
Value applied	0.75
Justification of choice of data or description of measurement methods and procedures applied	The project selects Option 1 :Use a default value to determine ϕ_y in the tool "Emissions from solid waste disposal sites" (version 08.0), and the project belongs to the situation of Application A: The project activity mitigates methane emissions from a specific existing SWDS, thus 0.75 is chosen for ϕ_y
Purpose of Data	Ex Ante baseline emission calculation
Comments	-

²² Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Ex ante baseline emission calculation
Comments	-

Data / Parameter	DOC _{f,y}
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of Data	Ex ante baseline emission calculation
Comments	-

Data / Parameter	MCF _y
Data unit	-
Description	Methane correction factor
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>

Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	The landfill is controlled, including the following activities: (i) cover material; (ii) mechanical compacting; (iii) levelling of the waste. Therefore, 1.0 for anaerobic managed solid waste disposal sites is recommended by “Emissions from solid waste disposal sites (Version 08.0).”
Purpose of Data	Baseline emission calculation
Comments	-

Data / Parameter	DOC _j
Data unit	-
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)</i>

Value applied	Waste type j	DOC _j (% wet waste)
	Wood and wood products	43
	Pulp, paper and cardboard (other than sludge)	40
	Food, food waste, beverages and tobacco (other than sludge)	15
	Textiles	24
	Garden, yard and park waste	20
	Glass, plastic, metal other inert waste	0

Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Ex ante Baseline emission calculation
Comments	-

Data / Parameter	k _j
Data unit	1/yr
Description	Decay rate for the waste type j

Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)					
Value applied	Waste type j		Boreal and Temperate (MAT $\leq$ 20 °C)		Tropical (MAT>20 °C)	
			Dry (MAP/PET<1)	Wet (MAP/PET >1)	Dry (MAP<1000mm)	Wet (MAP >1000mm)
	Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04	0.06	0.045	0.07
		Wood, wood products and straw	0.02	0.03	0.025	0.035
	Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.05	0.10	0.065	0.17
	Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.06	0.185	0.085	0.40
Justification of choice of data or description of measurement methods and procedures applied	Climatic conditions for Sinop where project site located in: MAT – mean annual temperature: 14.3 °C MAP – Mean annual precipitation: 689.3 mm PET – potential evapotranspiration: 150 mm Therefore, Boreal (MAT$\leq$20 °C) and wet (MAP/PET >1) conditions are observed in the project site. Sources: https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?k=undefined&m=SINOP https://www.mgm.gov.tr/arastirma/buharlasma.aspx?s=2020					
Purpose of Data	Ex ante Baseline emission calculation					
Comments						

Data / Parameter	EF _{grid,OM,y}
Data unit	tCO ₂ /MWh
Description	Operating margin CO ₂ emission factor for grid connected power generation in year y in Turkey
Source of data	Calculated by Turkish Ministry of Energy and Natural Resources. 23
Value applied	0.7424 tCO ₂ /MWh
Justification of choice of data or description of measurement methods and procedures applied	The data was obtained from Turkish Ministry of Energy and Natural Resources.
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the first crediting period.

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y
Source of data	Calculated by Turkish Ministry of Energy and Natural Resources. ²⁴ Also, “Tool to calculate the emission factor for an electricity system (Version 07.0)” is applied.
Value applied	0.5552 tCO ₂ /MWh
Justification of choice of data or description of measurement methods and procedures applied	The data was obtained from Turkish Ministry of Energy and Natural Resources. Also, “Tool to calculate the emission factor for an electricity system (Version 07.0)” is applied.
Purpose of Data	For the calculation of the Baseline Emission
Comments	This parameter is fixed ex-ante for the first crediting period.

²³

<https://enerji.gov.tr//Media/Dizin/EVCED/tr/ÇevreVeİklim/İklimDeğişikliği/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

²⁴

<https://enerji.gov.tr//Media/Dizin/EVCED/tr/ÇevreVeİklim/İklimDeğişikliği/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

Data / Parameter	Ru
Data unit	Pa.m ³ /kmol.K
Description	Universal ideal gases constant
Source of data	Tool to determine the mass flow of a greenhouse gas in a gaseous Stream (Tool 8 Ver 03.0)
Value applied	8,314
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Ex ante calculation of the Baseline Emission
Comments	-
Data / Parameter	MMi
Data unit	kg/kmol
Description	Molecular mass of greenhouse gas CH ₄
Source of data	Tool to determine the mass flow of a greenhouse gas in a gaseous stream
Value applied	16.04
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Ex ante calculation of the Baseline Emission
Comments	-
Data / Parameter	Pn
Data unit	Pa
Description	Total pressure at normal conditions
Source of data	Tool to determine the mass flow of a greenhouse gas in a gaseous stream
Value applied	101,325

Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Ex ante calculation of the Baseline Emission
Comments	-
Data / Parameter	Tn
Data unit	K
Description	Temperature at normal conditions
Source of data	Tool to determine the mass flow of a greenhouse gas in a gaseous stream
Value applied	273.15
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Ex ante calculation of the Baseline Emission
Comments	-
Data / Parameter	TDLy
Data unit	%
Description	Average technical transmission and distribution losses
Source of data	Obtained from https://webapi.teias.gov.tr/file/9248a331-e529-47ca-8c49-22f8705f5810?download
Value applied	10.0%
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh
Description	Quantity of net electricity generated using LFG by the project activity in year y
Source of data	<i>Electricity meter</i>
Description of measurement methods and procedures to be applied	The quantity of electricity consumed by the project plant will be monitored based on the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	-
Monitoring equipment	The quantity of electricity consumed by the project plant will be monitored based on the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate.
QA/QC procedures to be applied	The electricity meters are used to measure the net electricity exported to the grid and used for billing purposes. Therefore the meters are sealed by TEİAŞ (Turkish Electricity Transmission CO) and/or through protocols where and when appropriate. The grid company is responsible for maintenance and calibration of the device. In accordance with the “Regulation on “Measurement and Measuring Tools”, electricity meters have to be calibrated within a frequency of once per 10 years. The accuracy class of the device is 0.5s.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EC _{pj,y}
Data unit	MWh
Description	Quantity of electricity consumed by the project activity in year y
Source of data	Electricity meter

Description of measurement methods and procedures to be applied	The quantity of electricity consumed by the project plant will be monitored via the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	The electricity meters are used to measure the net electricity exported to the grid and used for billing purposes. Therefore the meters are sealed by TEİAŞ (Turkish Electricity Transmission CO). The grid company is responsible for maintenance and calibration of the device. In accordance with the “Regulation on “Measurement and Measuring Tools”, electricity meters have to be calibrated within a frequency of once per 10 years. The accuracy class of the device is at least 0.5s
Purpose of data	Calculation of project emissions
Calculation method	The quantity of electricity consumed by the project plant will be monitored based on the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate.
Comments	The quantity of electricity consumed is monitored by the same equipment, which monitors the net quantity of electricity generated by the project activity.

Data / Parameter	FCH ₄ ,sent flare,y
Data unit	tCH ₄ /y
Description	Quantity of methane in LFG sent to the flare in year y
Source of data	Calculated based on the flow of LFG and the concentration of methane in the LFG that will be sent to the flare(s)
Description of measurement methods and procedures to be applied	-Flow meter will be subject to regular (in accordance to the manufacturer) maintenance and testing to ensure accuracy.
Frequency of monitoring/recording	Continuously
Value applied	Will be monitored ex post

Monitoring equipment	Flow meter and gas analyzer. Data to be aggregated monthly and yearly. Measured by a flow meter and a gas analyser. Data to be aggregated monthly and yearly
QA/QC procedures to be applied	Flow meter will be subject to regular (in accordance to the manufacturer) maintenance and testing to ensure accuracy.
Purpose of data	Baseline emission calculation
Calculation method	-
Comments	-
Data / Parameter	Opj,h
Data unit	h
Description	Operation of the equipment that consumes the LFG
Source of data	Project participants
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Hourly
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Baseline emission calculation
Calculation method	-
Comments	For each equipment unit j using the LFG monitor that the plant is operating in hour h by the monitoring any one or more of the following three parameters: (a) Temperature. Determine the location for temperature measurements and minimum operational temperature based on manufacturer's specifications of the burning equipment. Document and justify the location and minimum threshold in the PD;

	(b) Flame. Flame detection system is used to ensure that the equipment is in operation; (c) Products generated. Monitor the generation of steam for the case of boilers and air-heaters and glass for the case of glass melting furnaces. This option is not applicable to brick kilns. $Op_{j,h}=0$ when: (a) One of more temperature measurements are missing or below the minimum threshold in hour h (instantaneous measurements are made at least every minute); (b) Flame is not detected continuously in hour h (instantaneous measurements are made at least every minute); (c) No products are generated in the hour h. Otherwise, $Op_{j,h}=1$
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Data / Parameter	$F_{CH_4,PJ,y}$
Data unit	m^3
Description	Methane captured and destroyed/gainfully used by the project activity in the year y
Source of data	<i>Flow meter</i>
Description of measurement methods and procedures to be applied	Calibration and frequency of calibration is according to manufacturer's specifications or applicable law and regulations.
Frequency of monitoring/recording	Continuous
Value applied	-
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Calibration and frequency of calibration is according to manufacturer's specifications or applicable law and regulations.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	The amount of landfill gas utilized shall be monitored ex post, using continuous flow meters.

Data / Parameter	F _{CH4,EL,y}
Data unit	tCH ₄ /year
Description	Amount of methane in LFG which is sent to the genset for electricity generation in year y
Source of data	Calculated based on the flow of LFG and the concentration of methane in the LFG that will be sent to the genset
Description of measurement methods and procedures to be applied	Flow meter will be subject to regular (in accordance to the manufacturer). Maintenance and testing to ensure accuracy.
Frequency of monitoring/recording	Continuously
Value applied	Will be monitored ex post
Monitoring equipment	Measured by a flow meter and a gas analyzer. Data to be aggregated monthly and yearly. Measured by a flow meter and a gas analyzer. Data to be aggregated monthly and yearly
QA/QC procedures to be applied	Flow meter will be subject to regular (in accordance to the manufacturer). Maintenance and testing to ensure accuracy.
Purpose of data	Calculation of baseline emissions (ex post)
Calculation method	Calculated based on the flow of LFG and the concentration of methane in the LFG that will be sent to the genset
Comments	-
Data / Parameter	W _{CH4}
Data unit	% (m ³ CH ₄ /m ³ LFG)
Description	Methane fraction in the landfill gas in the year y
Source of data	Measured by using a methane analyzer (measuring methane content directly)
Description of measurement methods and procedures to be applied	The gas analyzer will be subject to a regular maintenance and testing regime to ensure accuracy. The gas analyzer will be calibrated according to manufacturer's specifications.
Frequency of monitoring/recording	Continuously.

Value applied	50% (Will be monitored ex post)
Monitoring equipment	The gas analyzing system is a modular construction and designed for stationary operation for measuring directly the fraction of methane in the landfill gas. The gas analyzer provides three analogue signals, CH ₄ , CO ₂ and O ₂ . The values are measured continuously. The proportion of the data to be monitored is 100%.
QA/QC procedures to be applied	The gas analyzer will be subject to a regular maintenance and testing regime to ensure accuracy. The gas analyzer will be calibrated according to manufacturer's specifications.
Purpose of data	Calculation of baseline emissions (ex post)
Calculation method	-
Comments	Methane fraction of the landfill gas and LFG flow must be measured on the same basis (either wet or dry).
Data / Parameter	Tt
Data unit	K
Description	Temperature of the landfill gas
Source of data	Thermometer
Description of measurement methods and procedures to be applied	The device is subject to regular maintenance and testing regime to ensure accuracy. They will be periodically calibrated according to the manufacturer's recommendation by project participants.
Frequency of monitoring/recording	Continuous. The measurement interval will be equal to or more than one sampling each hour (average value in a time interval not greater than an hour will be used in the calculations of emission reductions). All the data will be aggregated hourly, daily, monthly and yearly.
Value applied	Will be monitored ex post
Monitoring equipment	Thermometer
QA/QC procedures to be applied	The device is subject to regular maintenance and testing regime to ensure accuracy. They will be periodically calibrated according to the manufacturer's recommendation by project participants.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

5.3 Monitoring Plan

Monitoring will be carried out following the procedures set by applicable methodology and all applicable tools of the above explained parameters. The monitoring consists of:

Responsibilities for monitoring

The project owner is responsible for the operation and maintenance of the landfill and the installed equipment. The project owner is also responsible for the administration of the data, setting up a VER team who will be responsible for monitoring all data required to estimate emission reductions.

Registration of the monitored data

The Sinop LFG VER-Team will be responsible for quality assurance and quality control of the monitoring equipment. The data measured by the monitoring equipment will be stored and will be processed into a monitoring report, which will be submitted by the project owner. All the monitored data will be stored during the crediting period and for at least two years after the end of the crediting period, whichever occurs later.

Corrective actions and emergency preparedness

The VER-Team will regularly check the monitoring system for errors. In the case of errors, corrective actions will be undertaken by the VER-Team, or if required, by the supplier of the monitoring equipment.

QA/QC procedure

Strong quality assurance and quality control procedures will be taken to monitor the equipment and data collection. Equipment and facilities will be subject to a regular maintenance and testing regime to ensure accuracy following the supplier's manual. In case of data deficiency or in correct data reading, the data will be crosschecked with other parameters and data leading to the most conservative emission reductions will be considered for the calculation of the emission reductions.

Measuring of exported and imported electricity

The main meter that monitors electricity import and export is placed at the grid connection. Monthly protocols based on the readings that are confirmed by both TEİAŞ and the project participant are prepared by the end of each month and stored. These protocols are also the basis of invoicing for the electricity sales from the proposed project activity. The electricity meters are bidirectional and will measure both the imported and exported electricity from and to the grid.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.2 Baseline Emissions

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.3 Project Emissions

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.4 Leakage

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.5 Net GHG Emission Reductions and Removals

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

APPENDIX X: